



Insights

100 AI Ideas You Can't Vibe-Code: Untapped Startup Opportunities for Expert Founders

James McClure, General Partner at Antler Australia, explores 100 unique, complex, and untapped AI startup ideas that can't be vibe-coded in a weekend - spanning sovereign GPUs, low-bandwidth AI, multi-agent systems, neurotech, synthetic audiences, and hypersonic space navigation. These high-barrier opportunities are launchpads for domain experts and builders in deep tech, regulated, scientific, or frontier domains. Read more below and join Antler's residency to build them now.



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EMERGING & UNEXPLORED

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Introduction to Edge AI

Antler is the world's most active AI investor, with notable investments including Lovable, Wrtn, Tactiq and Unseen. This means we see thousands of AI pitches each year.

I love an AI travel agent or AI SDR as much as anyone, but I'm looking for the cognoscenti or experts building in non-obvious spaces. The spaces that are being slept on, which only a few people in the world could build in, and which you definitely can't vibe-code in a weekend.

I'm particularly interested in the intersection between AI and the physical world, science, and secure or regulated industries.

As an investor who loves an analogy for each idea, I've chosen an "X for Y" comparison designed to be a launchpad for discussion.

Consider this an open invitation to riff on topics; the two best lines in any sparring session are "Tell me more" and "Yes, and building on that...".

And if you can build any of these, contact me [directly](#) or apply [here](#).



National Security & Sovereignty

1. AI-Powered Geopolitical Signals Engine

An AI model fine-tuned only on obscure forums, niche blogs and multilingual news to surface early indicators of geopolitical shifts for quant traders and political analysts.

[Stratfor](#) meets [Palantir](#).

2. Sovereign Dual-Use GPU Cloud for Australia

A sovereign GPU cloud designed to serve both defence and high-regulation industries like healthcare/mining/critical infrastructure, which is fully hosted in Australia, with export control compliance and air-gapped zones for sensitive AI workloads.

NextDC for AI compute, but only in Australia.

3. Secure AI Sandboxing Infrastructure

A Vercel-style deployment platform, but for AI models used in defence and other critical industries, with only pre-approved policies and protocols allowed.

AWS GovCloud, but with inference guardrails.



structured datasets and metadata.

ScreenApp for drone footage and metadata.

5. Satellite Payload Optimisation Copilot

A generative AI that designs, simulates and validates space payloads for commercial and defence missions.

Figma for SpaceX.

AI Infrastructure & Compute

6. Fine-Tune Marketplace Exchange

A protocol where creators can sell or license custom-trained models and fine-tune with embedded trust and provenance.

agent.ai for fine-tuned models.

7. AI Model Fragment Compiler

Break up LLMs into modular logic blocks that can be recomposed, like microservices.

Docker for LLMs.

8. Carbon-Aware AI Training Scheduler

A backend service that schedules AI workloads based on grid carbon intensity and GPU spot market availability.

A greener [Coreweave](https://coreweave.com), without the surge pricing.

9. Foundation Model Rollback Layer

An open-source version control for models, allowing safe rollback, audit trails, and reproduction of previous inference states.

[Git](https://git.com) for model state and memory.

10. Multi-Region AI Load Balancer



Edge AI & Low-Bandwidth Environments

11. Radio Frequency-Optimised LLM Decoder

LLMs adapted to interpret noisy radio signals, identify intent, and auto-correct errors in field transmissions.

Grammarly for fuzzy sound

12. Satellite-Deployable ML Training Pod

A deployable hardware + model bundle for training AI models on-the-fly in disconnected edge environments (e.g. low orbit or deep field).

Hugging Face in a box, but built for low/no bandwidth.

13. Offline Voice Assistant for Emergency Services and Diagnostics

A domain-specific agent deployed on handheld medical devices for triage, decision support, and record-keeping in disconnected settings.

Siri offline.

14. Power-Saving AI Chipsets

Model inference chips that can store and run on local solar, kinetic, or thermal energy sources

Toyota Prius meets NVIDIA.

15. Compressed LLM-as-Firmware

LLMs distilled to fit within the firmware layer of industrial machines, enabling local decision-making without middleware.

Samsung's Gauss, but for industrial machinery.



Regulated Industries & Compliance

16. Synthetic Audit Trail Generator



17. AI for Pharmaceutical Label Auditing

Parse, flag and compare discrepancies in global drug packaging/ warnings /labels using multilingual NLP trained on FDA, TGA, etc. documents.

Grammarly for pharma packaging and instructions (if it also saved you from major fines...)

18. Freedom of Information / GDPR Response Agent

Fine-tuned and secure LLM trained on internal email archives, contracts and CRMs to auto-generate legally compliant responses to Freedom of Information (FOIA), GDPR Data Subject Access Requests (DSAR) and subpoenas without breaching confidentiality or over-disclosing.

Ironclad for FOIA and DSAR requests.

19. Embedded RegTech for Web3 Buy-Side LLMs

An LLM layer that pre-screens DeFi trades, cross-chain asset flows, and jurisdictional exposures in real time to flagging sanctions risk, wallet blacklists, and regulatory boundaries before your bot executes.

Chainalysis meets 'GPT for crypto compliance.'

20. Real-Time Consent Agent for Healthcare Systems

An autonomous agent that dynamically manages patient consent across hospital systems, researchers, and third-party vendors using smart contracts.

OAuth for patient data and rights.



Multi-Agent Systems & Trust

21. Agent Reputation Protocol

Build an independent reputation layer for autonomous agents: track task success, collaboration scores, slippage, and prompt abuse history.

Trustpilot meets Experian for agents.

22. Multi-Agent Conflict Resolution Engine



Agency for AI Agents

23. Inter-Agent Memory Coherence Layer

Design shared memory infrastructure so agents working on separate tasks can synchronise learnings, context and strategies.

Databricks Lakehouse for intelligent collaboration.

24. Prompt Mutation Insurance for Agents

A cryptographic watermark and trace system that tracks prompt mutations across agent workflows and flags unintended drift.

Git for prompts.

25. Zero-Knowledge Proof Layer for Agents

Before a model acts, it proves it knows what it's doing, but without exposing how to provide the backbone of AI in zero-trust environments.

Proof-of-Work for AI decisions.



GovTech & Legacy Systems

26. COBOL Copilot for Government Systems

An LLM fine-tuned on legacy programming languages like COBOL, FORTRAN, etc., built to maintain, refactor or translate ancient codebases still running governments and banks

Cursor for crap code.

27. Automated Legislation Harmonisation Engine

A model trained to reconcile overlapping federal and state laws: highlight contradictions, suggest resolutions and track legislative deltas.

Stack Overflow for lawmakers.

28. Spreadsheet Agent for Municipal Utilities

An AI layer that lives inside legacy Excel sheets used by water, electricity and transport departments, which map, debug and simulate operations without requiring a full



29. AI Retrofit Planner for Ageing Buildings

A computer vision + code interpreter system that ingests floor plans, sensor data and energy usage to recommend and simulate retrofit strategies for buildings built before 1980.

Databricks for old buildings.

30. Industrial Machine Predictive Analytics without IoT or API

A vibration/sound-based agent that learns the acoustic signature of ageing machinery (factories, pumps, engines) and flags failure risk or tuning needs without IoT or data feeds from the machines.

Factory AI for analogue machines.

Synthetic Data & Digital Verification

31. Deepfake Consistency Checker for Newsrooms

A model trained to cross-reference image metadata, lighting patterns and historical footage to detect AI-manipulated media.

Snyk for newsroom content integrity.

32. Synthetic Transaction Generator for Fraud Testing

A platform that simulates realistic, adversarial banking transactions to stress-test fraud detection systems.

Penetration testing for fraud detection.

33. Reverse-Prompt Forensics Engine

Given an image, document, or audio, infer the likely prompts or instructions that created it.

Shazam for AI content.

34. Digital Twin Validator for Industrial AI



Antler for Startups & Founders

35. Open Source Registry of Synthetic Datasets

A version-controlled, peer-reviewed repository of synthetic datasets for training regulated-domain models, including lineage, use-case tags and risk scores.

Hugging Face for regulated synthetic data.



Neurotech & Human-Machine Interfaces

36. Brain-First Model Tuning Toolkit

A platform that adapts LLM response latency, tone, and memory depth based on EEG or neural signal feedback.

OpenAI meets Neuralink.

37. Emotion-to-Interface Modulator

A system that interprets real-time emotional states (via EEG, facial micro-expressions or HRV) and adjusts the UI/UX of software interfaces based on the user's psychological state.

Figma for dynamic emotion-aware UX.

38. Adaptive Attention Training via AI Gamification

AI-generated cognitive tasks that evolve in real time based on neurofeedback and performance, designed to train focus, working memory, or creativity.

Duolingo for focus and neuroplasticity.

39. Silent Speech Interface for Workspaces

Using subvocalisation and biosignal inference, let users 'speak' to their AI assistant without saying a word - ideal for secure or shared environments.

Whisper AI + Tactiq for thoughts (and maybe used by spies).

40. Cognitive Load Balancer for AI Agents

Monitors human fatigue, overload or stress via wearable signals and adapts AI responses accordingly to pause, simplify or delay interactions.



41. Synthetic Fandom Communities

AI agents trained to mimic superfans of franchises or creators to simulate, grow, and moderate niche communities.

Reddit moderators as agents.

42. Synthetic Audience Simulators for Creative Media

Train LLMs on audience/critic data to simulate emotional and behavioural responses to scripts, trailers or new music prior to launch - replacing 'rushes' or 'album listening parties'.

[Usertesting.com](https://www.usertesting.com) for creative content.

43. Synthetic News Personalities

Create synthetic anchors/ columnists/ bloggers with custom tone, bias and energy to deliver news tailored to regional/ ideological/ niche interests.

Russia Today, but personalised to an audience of dozens, not millions.

44. Synthetic Polling Data

Train a synthetic population model (built from census data, social media embeddings, consumer segments, and mobility patterns) to simulate representative polling responses, offsetting the challenge of building a representative sample.

YouGov without human bias (but with AI bias).

45. Synthetic Ad Ratings Engine

An AI platform that simulates how diverse population segments react to advertising creative, before anything is launched.

Nielsen for synthetic audiences.



AI-Native Marketplaces

46. Model-to-Model Freelance Marketplace



more relevant

47. Synthetic Dataset Exchange for Industry Verticals

A licensed B2B marketplace where providers upload, monetise, and benchmark synthetic data for domains like legal, oil & gas, and insurance.

OpenSea for synthetic data.

48. Encrypted Pool for Confidential Model Training

Private, encrypted training venues for corporates to fine-tune on sensitive data without leakage.

Signal for Corporate LLM models.

49. Prompt Arbitrage Network

A decentralised system where agents route tasks to the best model + prompt combo for cost, latency, or performance.

Uber Freight for inference.

50. Insurance Exchange for AI Model Risks

A regulated marketplace that underwrites, packages, and prices model performance risk across verticals.

Upcover for Agent Liability.



Bio + Health AI

51. Adaptive Drug Dosing Copilot

An AI that dynamically recommends dosage adjustments based on real-time patient biomarkers, wearable data and drug interaction risk.

Lovable for prescriptions and dosing.

52. Protein Folding Validator Using Foundation Models

An adversarial LLM that predicts folding anomalies, flags hallucinations and benchmarks AI-generated proteins before wet lab spend.



Train a generative model to suggest hypothetical - but biologically plausible - paths for rare disease aetiology and generate candidate gene targets.

TensorFlow for rare diseases.

54. Bio-Robotic Feedback Trainer

A startup building a real-time ML control loop between bio-printed tissues and robotic manipulators, which teaches soft tissue to "think" like muscle.

Waymo for soft robotics.

55. Molecular Copilot for Lab Scientists

An agent that predicts next experiments, flags protocol errors and books the High Performance Liquid Chromatography before you've even finished your coffee.

Linear for wet labs.

56. AI-Native Clinical Trial Designer

Generate, simulate and optimise clinical trial structures using population-level synthetic data and historical outcome benchmarks.

Figma meets Airtable for clinical trials.

57. Cellular Agriculture Automation Agent

An AI stack that handles feeding schedules, cell environment adjustments and failure detection in cultured meat bioreactors and similar production lines.

Datadog + agents for cell cultures.

58. Wearable-Free Glucose Trend Predictor

Use phone sensors, sleep, diet logs, and ambient cues to predict glucose spikes without a Continuous Glucose Monitor.

Virtual Whoop for CGM.

59. Privacy-Preserving AI for Genomic Sequencing



Empowering the generation

60. Voice-Based Mental Health Biomarker Detector

AI that flags cognitive fatigue, depression risk or neurodegeneration from voice tone, cadence and sentiment over time.

Oura for mental health and integrated with pre-existing hardware.



61. AI Architect for Orbital Infrastructure

An agent trained on orbital mechanics and propulsion physics, which is capable of auto-designing satellite clusters and repair drones.

Figma with an Orbital Infrastructure PhD.

62. Inference Engine for Hypersonic Navigation

An edge model that fuses radar, inertial sensors, and atmospheric data without GPS dependence.

Tesla Autopilot at Mach 5.

63. Dream-to-3D Generator

Fine-tuned multi-modal model that converts EEG or sleep-stage patterns into low-res 3D scene approximations.

Veo 3.0 for dreams.

64. Volcanic Eruption Prediction with Multimodal AI

Train a model on seismic activity, gas composition, satellite thermal imaging and historical eruption sequences to sell to real estate, hedge funds and insurance companies.

Scale AI for seismic activity.

65. Quantum-Safe Agent Communications

A startup building encrypted agent-to-agent prompt and messaging protocols with quantum security baked in.



Contact

Train AI to simulate plausible alien linguistic systems based on cognition models (and maybe some Star Wars scripts).

Duolingo meets SETI.

67. Terraforming Simulation AI

A multi-agent physics-informed simulation to explore plausible terraforming pathways across atmospheres, soil compositions and micro-ecosystems.

Unity for terraforming.

68. LLM for Long-Term Climate Strategy

Not carbon offsets, but rather actual multi-decade policy modelling with probabilistic impact forecasts for major carbon-producing companies.

SimCity meets AI simulator for corporate climate strategy.

69. Subsurface Mapping Copilot for Ocean Exploration

An AI that processes sonar, magnetometer and fluid flow data to generate real-time seabed maps for undersea mining or archaeology.

Matterport for the ocean floor.

70. Cold Atom Simulator for Quantum Lab Planning

Use LLMs fine-tuned on physics papers and simulation data to test cold atom configurations before they're ever built.

Vercel DX Platform for quantum labs.



AI for Physical World Interfaces

71. Structural Integrity Monitor Agent

AI that uses vibration, sound and visual data to detect bridge stress, pipe failure or foundation cracks in real-time - deployed on transport infrastructure

Factory AI for major infrastructure.



sensors are degrading and when to replace them.

New Relic for sensors.

73. Facility Energy Optimisation Agent

Ingests HVAC data, occupancy patterns and grid pricing to tune energy consumption - proactively or autonomously.

Nest for factories.

74. Latent Leak Detection System

Uses sonic anomalies, micropressure shifts and temperature fluctuations to detect leaks in gas, water or chemical pipelines

SiteHive for liquids.

75. AI Material Compatibility Validator

Computer vision + sensor stack, which verifies material quality and compatibility at delivery or receiving docks, which flags incorrect shipments or risky compositions

Top Boy's Sully for legitimate purchases.



Creator Economy and Solo Capitalism

76. Synthetic Persona Licensing Market

A platform where influencers and niche creators license synthetic versions of their voice, face, or persona to brands or fans, with programmable restrictions and rev-share baked in.

IMG meets HeyGen.

77. Micro-IP Generator for Fan Entrepreneurs

Tool that helps niche creators spin off character universes, merch and lore packs for their fandom to remix and resell - creating long-tail revenue via AI-generated IP kits

Patreon meets Canva for creator ecosystems.

78. Dynamic Micro-Storefront Generator



...operating... ...operating... ...operating...

79. Solo Operator Equity Tokeniser

An on-chain tool for solo creators and operators to tokenise future revenue streams (e.g. royalties, SaaS MRR, Substack revenue) and sell slices as micro-equity to fans or collaborators
Carta for your future earnings.

80. Ambient Tipping Network

A protocol that lets users tip anonymous contributors or avatars in public spaces (forums, design systems, Discord servers) - with attribution and payment handled on-chain.
Patreon for anonymity.



Science Breakthroughs and Phenomena

81. Hypothesis Blender for Longshot Science

An LLM trained on retracted papers, fringe theory forums and obscure preprints - designed to generate “weird but plausible” hypotheses to feed back into discovery loops
Ancient Apocalypse for scientific breakthroughs.

82. Anomaly Miner for Lab Instruments

Use AI to crawl ignored sensor data, logs, failed experiments or control group variance in labs. Not to confirm hypotheses, but to find weird patterns nobody expected.
Palantir for lab noise.

83. Emergence Detector for Complex Systems

AI model that scans real-time or historical data across biological, ecological and computational systems to flag moments of emergence (e.g. tipping points, phase transitions) and can predict future changes.
Grafana for chaos theory.

84. Paradigm Shift Signal Miner



85. Causal Inference Engine for Unobservable Phenomena

AI that proposes plausible causal graphs in domains with poor data visibility (e.g. social contagion, very early medical trials or multi-generational placebo effects). Not providing empirical truth, but structured hypothesis-generation in epistemically fuzzy terrain.

GraphQL for data lacking obvious correlation.

Cyberpunk Enterprise

86. AnonOps Dashboard

A zero-identity enterprise platform where agents operate with cryptographic identity only. No emails or names, just wallet-bound access. Ideal for high-stakes ops, crypto DAOs and whistleblower orgs.

Signal meets Relevance AI.

87. Fully Autonomous Shadow IT Mapper

Deploys LLMs across internal comms, browser extensions and endpoint behaviours to map every unauthorised tool, rogue Zap and workaround in your enterprise.

Vanta for the unofficial workflows.

88. Synthetic Employee Audit Agent

Create digital twins of employees with varying levels of compliance, knowledge or social engineering skill and insert them into enterprise workflows to test policy, ethics and security risk.

Sync Technologies for compliance and cyber.

89. Ephemeral Document Lifecycle Manager

Certain categories of internal documents self-expire, fragment or are deleted after a pre-defined number of views, contexts or agent interaction thresholds. Ideal for hyper-volatile workflows and deniable ops.



90. Cross-City Alliance Mapper

Maps entangled SaaS usage, cross-domain logins, joint ventures, ex-employee influence and backdoor connections across companies in a supply chain to enable better multithreading

Outreach augmented with off-market signals.



Advertising and Ecommerce

91. Reputation-Augmented Pricing Engine

Prices products based on who you are, not just availability. Discounts, bundles, or surcharges shift based on verified trust, prior behaviour or network influence.

Surge pricing but for customer quality.

92. Conversion Path Visualiser

Displays real-time generative diagrams of likely conversion paths based on real-time ad data

Cardlytics for GA360.

93. Incognito and Multi-Touch Attribution

Rebuild user paths across devices via contextual fingerprinting and their semantic behaviour patterns to fine-tune ROAS calculations

Mixpanel meets context windows.

94. Adspend Reconciliation Protocol

A shared, trustless ledger where advertisers and publishers reconcile campaign spend, delivery and outcomes - without arguing over click counts.

The Trade Desk decentralised.

95. Prompt-Level Sponsorship Exchange

A real-time bidding platform for embedding brand-sponsored prompts, tools or agents inside AI apps (e.g. "summarise this with Grammarly AI" or "analyse this contract with Lexis AI")

AdSense for LLM Prompts.



96. Dynamic Bill Of Materials Optimiser

A predictive engine that dynamically recalculates bill-of-materials based on commodity price shifts, supplier availability and real-time demand forecasts

MosaicML and commodities day trading for the supply chain.

97. Budgeting Scenario Planner based on live conversations

Mine email and Slack convos to predict future expenditure with probabilistic scenario planning.

Clari for opex.

98. Logistics Disruption Early Warning System

Model trained on shipping lanes, weather and ports data to predict disruptions in advance and reroute or manage inventory.

PredictHQ for freight.

99. Adaptive Warehouse Layout Optimiser

Software layer which redesigns warehouse layouts based on SKUs, pick times and seasonal velocity. Tested via simulation before real-world deployment.

Unreal Engine for pallets.

100. FX Day Trading Layer for Accounts Receivable

An AI-native accounts receivable platform that predicts FX movements and dynamically optimises invoice collection, currency conversion and settlement timing to maximise cash value.

[Tradersuite.ai](https://www.tradersuite.ai) for unpaid invoices.

Ready to build?

MORE *INSIGHTS*



SPRING 2026 U.S. PORTFOLIO *SHOWCASE*



Insights April 8, 2026



Announcing the Spring 2026 U.S. Portfolio Showcase

Meet the portfolio companies featured in our Antler U.S. Spring 2026 Showcase.

The Tipping Point for *PHYSICAL AI*



Insights March 30, 2026



The Physical AI Inflection Point

Why we believe the moment for Physical AI and Robotics has finally arrived, our early thinking about the opportunity, and where we are placing our bets.



Insights March 26, 2026



Software Was Never About the Code

Software encodes process. AI changes who can do the encoding but it doesn't change the need for someone to know what should be encoded.



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